

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DevOps Lab Experiment – 8

Experiment 8: Practical Exercise: Set Up a Jenkins CI Pipeline for a Maven Project, Use Ansible to Deploy Artifacts Generated by Jenkins.

1. Create a Maven Project

Open IntelliJ IDEA IDE → Click on **New Project** → In the left panel, select **Maven Archetype**
Enter the project details as shown below:

Name : HelloMaven

Location : Choose the location where you want to save your project.

JDK : 25 or any installed version

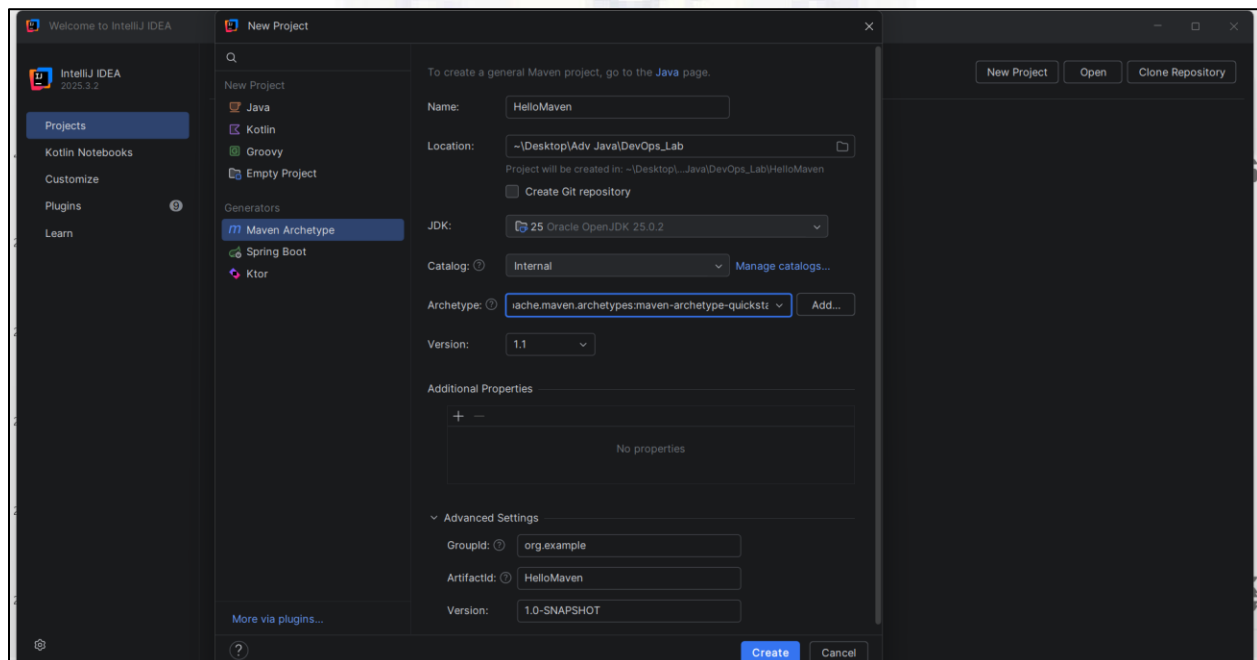
Archetype : org.apache.maven.archetypes:maven-archetype-quickstart

Version: 1.1

Configure:

- GroupId : org.example (or com.example)
- ArtifactId : HelloMaven
- Version : 1.0-SNAPSHOT

Click **Create**



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Add the **<build>** section in **pom.xml** file

```
<build>
<plugins>
<plugin>
  <groupId>org.apache.maven.plugins</groupId>
  <artifactId>maven-jar-plugin</artifactId>
  <version>3.3.0</version>
  <configuration>
    <archive>
      <manifest>
        <mainClass>org.example.App</mainClass>
      </manifest>
    </archive>
  </configuration>
</plugin>
</plugins>
</build>
```

Build and execute the jar file

mvn clean package

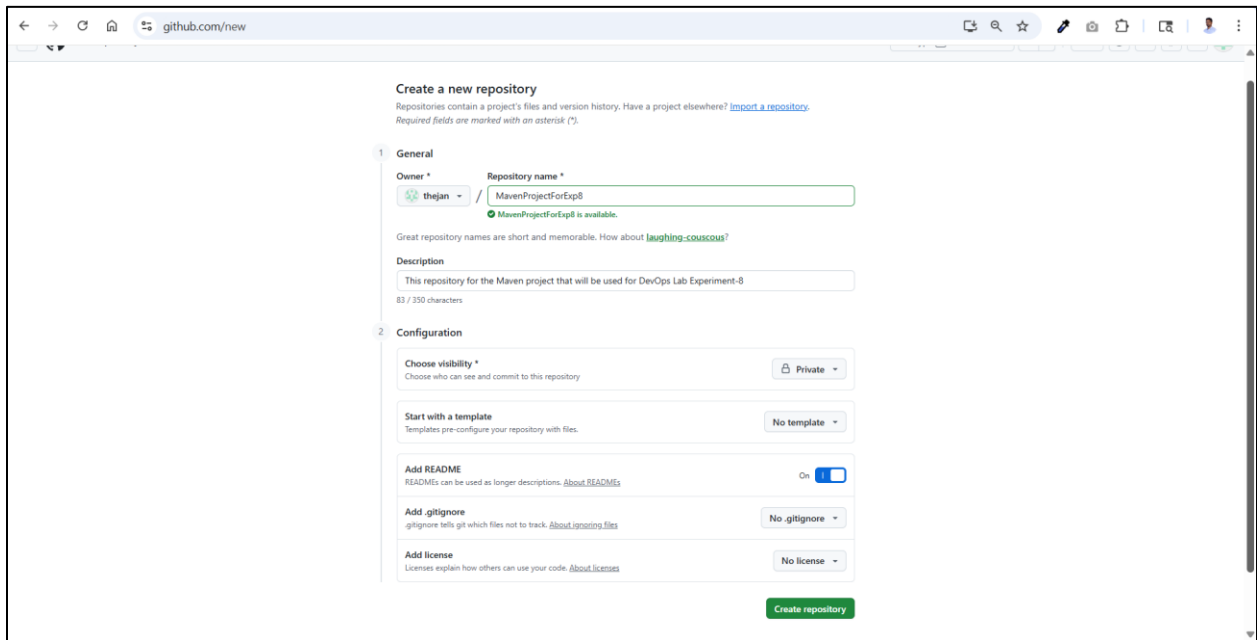
java -jar target/hellomaven-1.0-snapshot.jar

```
PS C:\Users\thejan\Desktop\Adv Java\DevOps_Lab\HelloMaven> java -jar target/HelloMaven-1.0-SNAPSHOT.jar
Hello World!
```

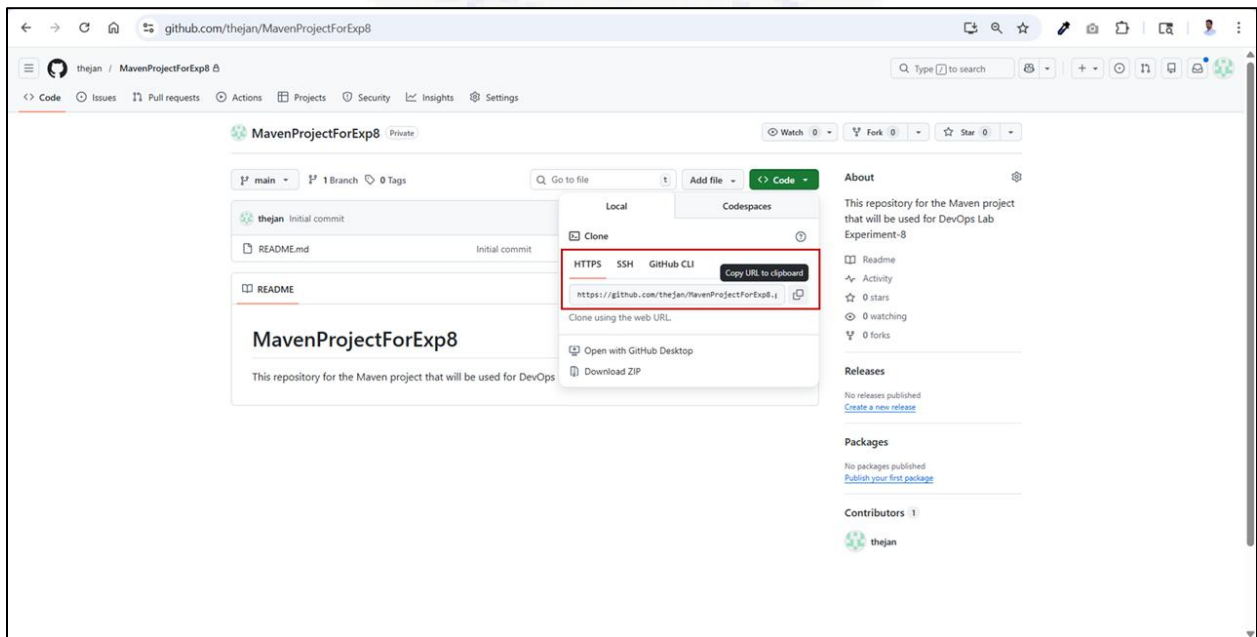
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2. Create a repository on GitHub and push the HelloMaven project

Log into GitHub account and create a new repository and copy the repository URL as show in below figures



Creation of new repository

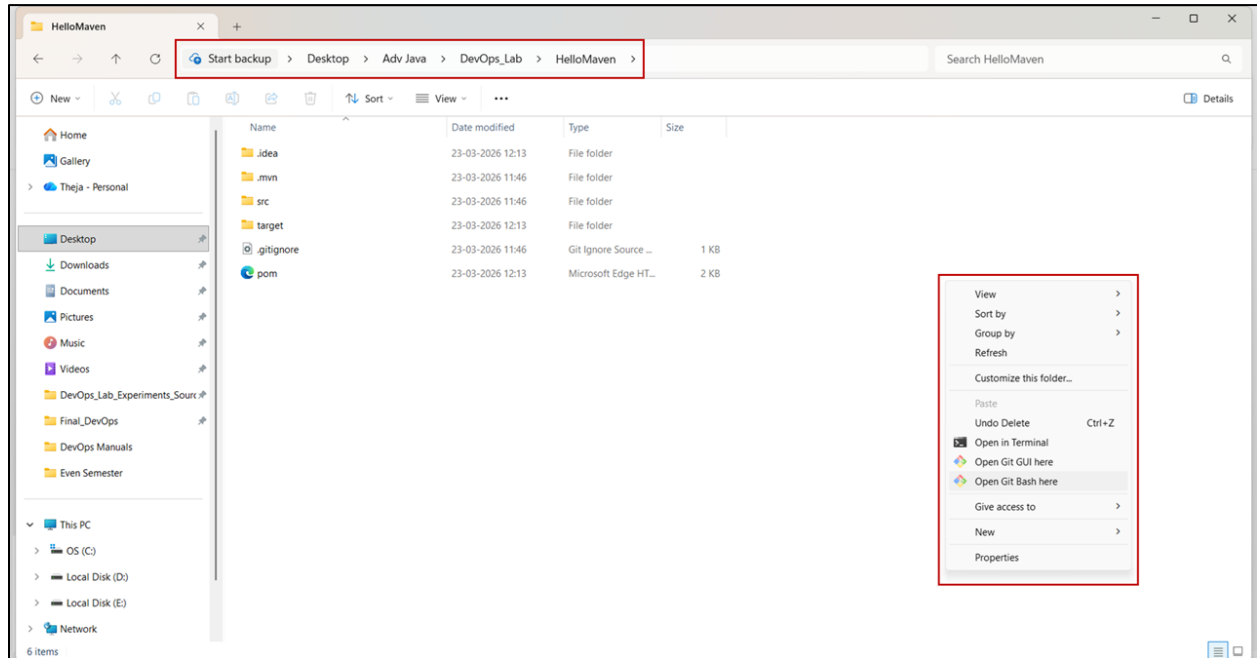


Copying Git repository URL

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Rename the main branch to master in the remote Git repository.

Navigate to the project directory, open Git Bash terminal and execute the git commands



git init

git status

git add .

git remote add origin <URL_of_your_git_repository>

git commit -m "Initial Commit of HelloMaven Project"

git pull origin master --allow-unrelated-histories (if required)

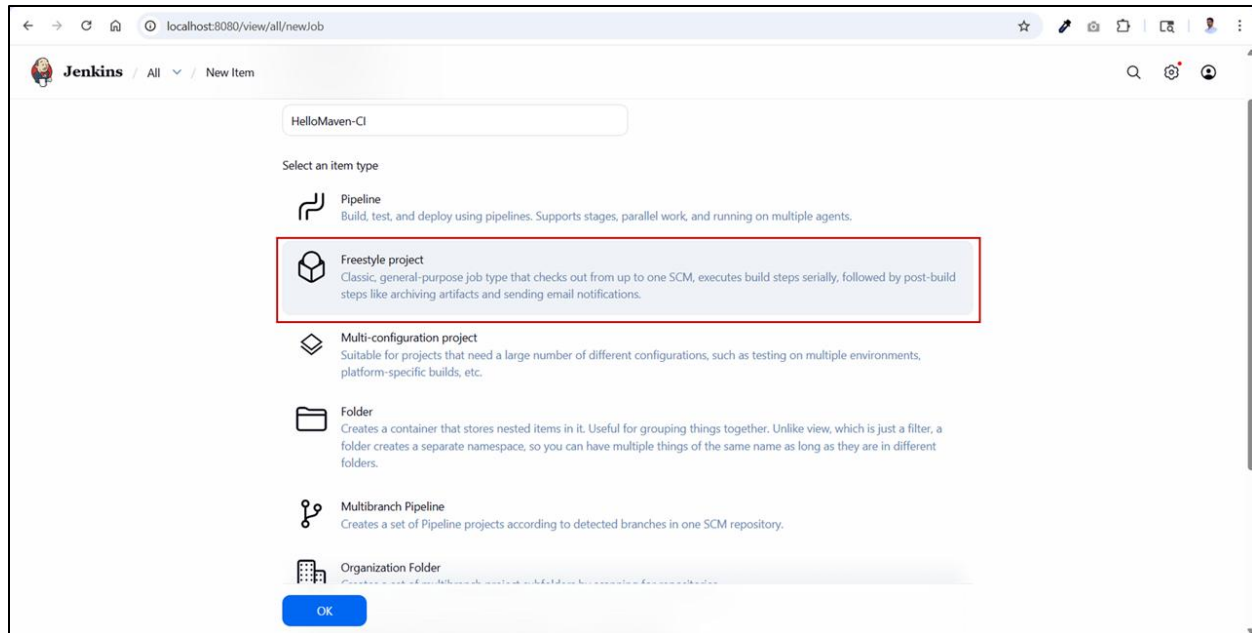
git push -u origin master

3. Configuring Jenkins to Build the Maven Project

Start the Jenkins server using the command `java -jar jenkins.war`. Then, open `http://localhost:8080/` in a web browser and log in to the Jenkins account.

Click on **New Item**, enter the name as **HelloMaven-CI**, select **Freestyle project** and click on **OK**

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Enter the description as **CI pipeline for HelloMaven project**

Configure Source Code Management

Select Git

Repository URL : Enter the repository URL

Credentials : If the repository is private, click “Add” and provide the necessary credentials

Branch Specifier : */main or */master

Repository browser : auto

Build Steps

Click on + **Add build step** and select Execute Windows batch command

Command : **mvn clean package**

Click again on + **Add build step** and select Execute Windows batch command

Command : **ws1 ansible-playbook -i inventory.ini deploy.yml**

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Post build Actions

Click on **+Add post-build action** and select **Archive the artifacts**

Files to archive : target/*.jar

This pattern tells Jenkins to archive any JAR file found in the target directory

And click on **Save** button to save the configuration.

4. Writing an Ansible Playbook for Deployment

Create an Ansible playbook that deploys the Maven artifact (the JAR file) generated by Jenkins to a target directory.

- Navigate to the project directory → Right click inside the blank space → Select **Open in Terminal**
- Create an inventory file : **ni hosts.ini** and add the following content to it:

```
local]
localhost ansible_connection=local
```

hosts.ini

- Create playbook file : **ni deploy.yml** and add the following content to it.

```
---
- name: Deploy Maven Artifact (Jenkins User Accessible)
  hosts: localhost
  become: yes

  tasks:
    - name: Ensure deployment directory exists
      ansible.builtin.file:
        path: /home/thejan/deployment
        state: directory
        mode: '0755'

    - name: Copy the artifact to the deployment directory
      ansible.builtin.copy:
        src: /var/lib/jenkins/workspace/HelloMaven-CI/target/HelloMaven-1.0-SNAPSHOT.jar
        dest: /home/thejan/deployment/HelloMaven.jar
        remote_src: yes
```

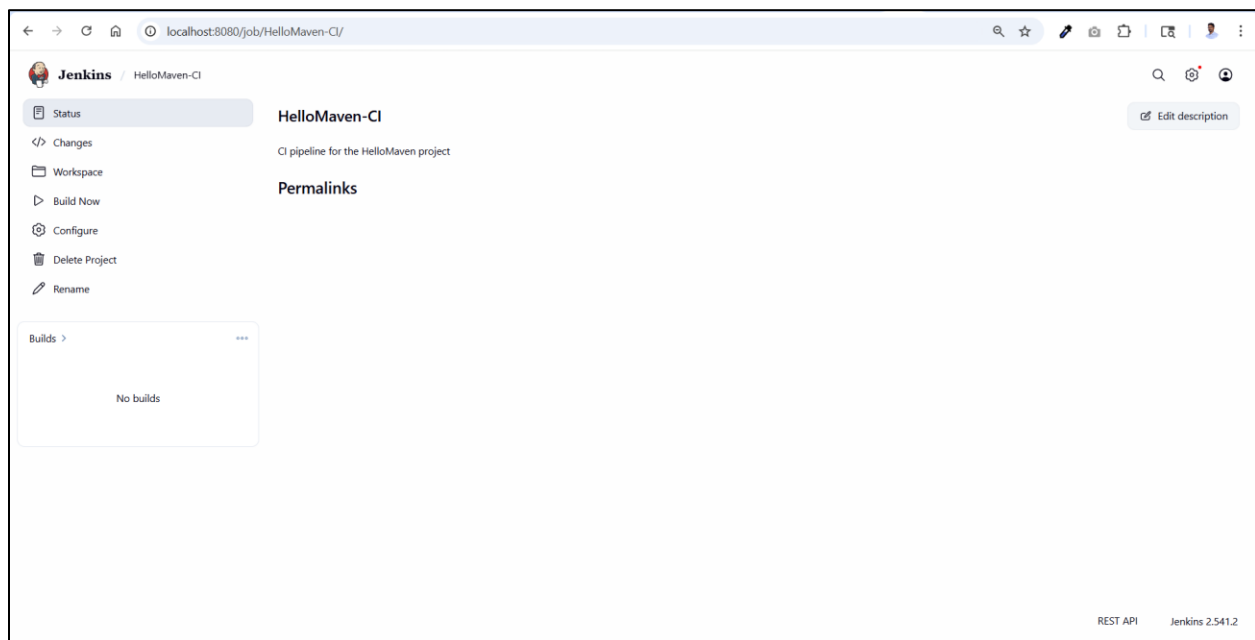
deploy.yml

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- Add these **ini** and **yaml** files to the git repository
git add .
git commit -m "Added inventory and playbook files"
git push -u origin master

5. Testing the complete pipeline

Navigate to Jenkins job page → Click on **Build Now**



Once the build completes, click the build number (e.g., #1) and check the **Console Output**.

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```
PLAY [Deploy Maven Artifact (Jenkins User Accessible)] *****

TASK [Gathering Facts] *****
1 ok: [localhost] System information (facts) was gathered successfully

TASK [Ensure deployment directory exists] *****
2 ok: [localhost] Directory already exists OR created successfully.

TASK [Copy the artifact to the deployment directory] *****
3 changed: [localhost] File was copied OR updated successfully.

PLAY RECAP *****
localhost : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

C:\Users\thejan\.jenkins\workspace\HelloMaven-CI>exit 0
Archiving artifacts
Finished: SUCCESS
```

The Maven artifact was successfully deployed to the target directory using Ansible

Verify that the deployment folder contains the deployed JAR file.

```
Windows PowerShell
thejan@ThejaNarayana: ~/de
thejan@ThejaNarayana: ~/deployment$ cd /home/thejan/deployment
thejan@ThejaNarayana: ~/deployment$ ls
HelloMaven.jar
thejan@ThejaNarayana: ~/deployment$
```

VIVA Questions

6. What is the role of Jenkins in this experiment?
7. What is Maven and why is it used?
8. What is the purpose of the pom.xml file?
9. What is the function of mvn clean package command?
10. What is a JAR file?
11. What is the role of GitHub in this pipeline?
12. What is a repository in Git?
13. What is Jenkins pipeline?
14. What is a Freestyle project in Jenkins?
15. What is the purpose of “Build Steps” in Jenkins?
16. What is the purpose of Archive the artifacts in Jenkins?

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17. What does target/*.jar mean?
18. What is Ansible and why is it used?
19. What is an Ansible playbook?
20. What is an inventory file in Ansible?
21. What is the meaning of localhost ansible_connection=local?
22. What is the purpose of become: yes in Ansible?
23. What does the file module do in Ansible?
24. What does the copy module do in Ansible?
25. What is the meaning of remote_src: yes?
26. What is the difference between ok and changed in Ansible output?
27. What does failed=0 indicate in the play recap?
28. What is the purpose of “Gathering Facts” in Ansible?
29. How does Jenkins trigger the Ansible playbook?
30. Why do we use WSL for running Ansible in Windows?
31. What is the workflow of this complete pipeline?
32. How do you verify that deployment is successful?
33. What is the difference between build and deployment?
34. Can Ansible deploy to remote servers?
35. What are the advantages of using Jenkins with Ansible?
36. What is the final outcome of this experiment?